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Review

Flexible fixation versus open reduction internal fixation and primary arthrodesis for ligamentous Lisfranc injuries: A systematic review and meta-analysis

Kyle P. O'Connor [a], Erica R. Olfson [b], John T. Riehl [a,*]

[a] Med City UNT/TCU Orthopaedic Surgery Residency Program, Denton, TX, USA
[b] Texas Christian University, Burnett School of Medicine, Fort Worth, TX, USA

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ABSTRACT

Introduction: Flexible fixation (FF) has allowed treatment of isolated ligamentous Lisfranc injuries while preserving joint motion. We hypothesize that patient-reported outcome measures (PROMs), complications, and return-to-activity rates will be similar between patients undergoing FF versus those undergoing open reduction internal fixation (ORIF) or primary arthrodesis (PA).
Methods: Databases included PubMed, OVID Medline, Embase, SCOPUS, Cochrane Central Register of Clinical Trials, and clinicaltrials.gov from their inception to 5/13/2024. Search terms focused on treatment of Lisfranc injuries with FF, ORIF, or PA. Only English studies were included. Studies were included if the Lisfranc injury was purely ligamentous and had PROM scores. Quality, validity, and comparability were assessed using MINORS and GRADE criteria. Meta-analysis was conducted using pooled statistics. Cohen's d and odds ratios (OR) determined effect sizes.
Results: Twenty-five studies were included. There were 184 patients undergoing FF, 236 patients undergoing ORIF, and 80 patients undergoing PA. Postoperatively, American Orthopaedic Foot and Ankle Society (AOFAS) scores were 89.7 ± 10.0, 78.7 ± 44.2, and 87.4 ± 31.8, VAS-pain scores were 1.5 ± 1.5, 1.6 ± 3.8, and 0.3 ± 2.6, and return to activity rates (RTA) were 100 %, 63.3 %, and 78.4 %, respectively. Rates of post-traumatic arthritis were 0 %, 13.0 %, and 0 %, hardware removal were 0 %, 86.0 %, and 22.5 %, and complications were 3.8 %, 17.7 %, and 23.5 %. Meta-analysis demonstrated that FF had superiority over ORIF regarding better AOFAS scores and RTA with lower rates of post-traumatic arthritis, hardware removal, and complications ($p < 0.05$). Also, FF had superiority over PA with higher RTA and lower rates of hardware removal and complications. PA demonstrated better VAS-pain scores ($p < 0.05$).
Conclusion: FF had satisfactory outcomes after Lisfranc injury treatment. Low-quality evidence suggested that FF had better outcomes, however, this conclusion was drawn from single-arm studies which have significant limitations. Further prospective, comparative studies should investigate this relationship.

1. Introduction

The Lisfranc ligament connects the medical cuneiform to the second metatarsal base and stabilizes the medial column which maintains the midfoot arch. [1] Injury occurs with a rotational and axial force through a hyper-plantarflexed foot which causes diastasis between the first two metatarsals. [2] The most established forms of treatment include open reduction internal fixation (ORIF) and primary arthrodesis (PA) to help stabilize this joint space and decrease the chance of chronic pain, deformity, and dysfunction. [3] While ORIF and PA have demonstrated effectiveness in improving patient-reported outcome measures (PROMs), there have been associations with articular damage, hardware failure, and need for eventual hardware removal with these treatment modalities [4,5].

An alternative to ORIF and PA involves the use of flexible fixation (FF) devices to reconstruct Lisfranc ligaments for isolated ligamentous injuries.[6] Potential benefits include minimized trauma to the articular cartilage and near elimination of the need for hardware removal after recovery.[7] Furthermore, FF more closely recreates the physiology of the Lisfranc joint by allowing movement. Currently, most published

* Correspondence to: Med City UNT/TCU Orthopaedic Surgery, 3535 S Interstate 35, Denton, TX 76210, USA
E-mail address: jtriehl@hotmail.com (J.T. Riehl).

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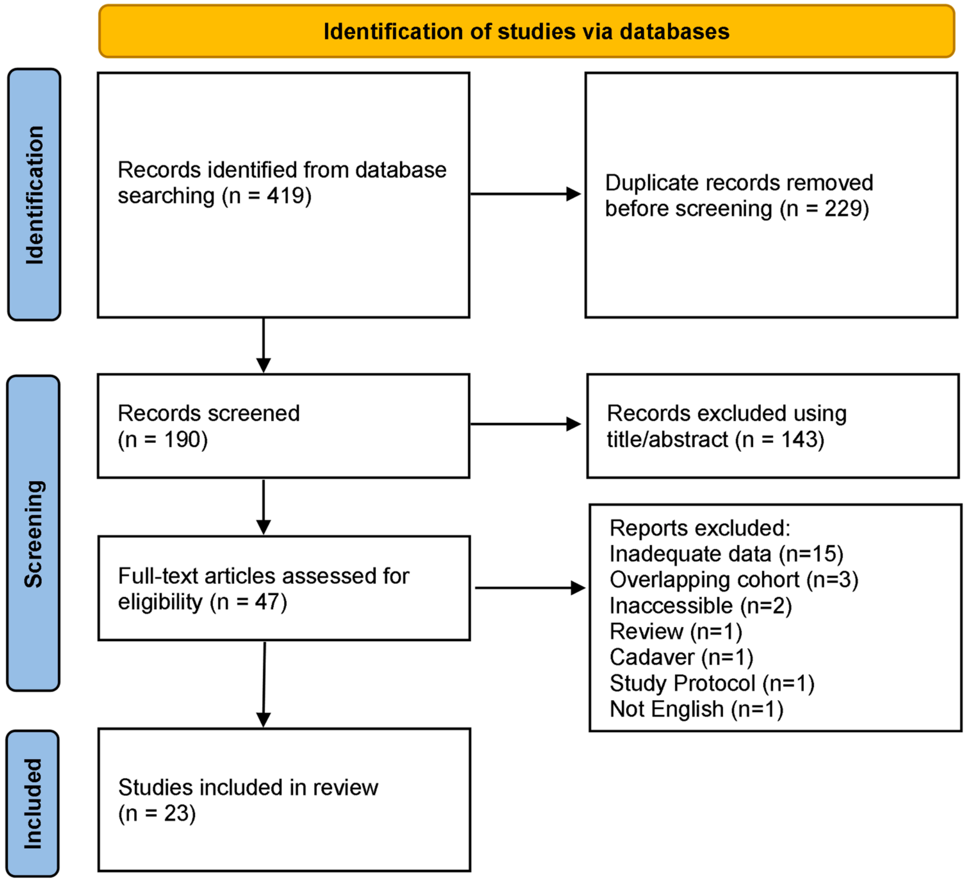

Fig. 1. PRISMA diagram.

studies are single arm with small cohort sizes. The two available comparative studies between FF and screws demonstrated no difference in outcomes, although these studies have limited power [8,9].

This systematic review and meta-analysis (SRMA) provided a comprehensive evaluation of the entire Lisfranc FF literature and the first meta-analysis using comparable control arms of ORIF and PA patients that underwent treatment for isolated Lisfranc ligamentous injuries.

2. Methods

2.1. Study design

To determine a clinical question, the PICO(T) (Population, Intervention, Comparison or control, Outcome, and Type of Study) model was used with the intervention/therapy question type. [10] The PICO(T) question was "In patients with Lisfranc injuries, how does FF (intervention) compared with ORIF and PA (comparison) affect patient-reported outcome measures (PROMs) and return to activity (RTA) (outcome) within single- and double-arm studies (type of study). The Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines were followed. [11] Databases searched were PubMed, OVID Medline, Embase, SCOPUS, Cochrane Central Register of Clinical Trials, and clinicaltrials.gov from their dates of origin to 5/13/2024. Search terms included Lisfranc, metatarsal fracture, metatarsal dislocation, tarsometatarsal, flexible fixation, suture button, suture anchor, InternalBrace, ligamentous, open reduction internal fixation, and primary arthrodesis.

2.2. Inclusion and exclusion

Study inclusion was conducted by two independent authors (K.P.O. and E.R.O.). Inclusion criteria were studies with (1) purely ligamentous Lisfranc injuries, (2) clinical outcomes (PROMs or RTA) for FF, ORIF, or PA, and (3) mean 1-year follow-up duration. Exclusion criteria included reports that did not satisfy all inclusion criteria, non-English studies, basic science studies, animal studies, cadaver studies, technique papers, case reports, conference abstracts, editorials, letters to the editor, supplements, and guidelines. Overlapping cohorts were excluded.

2.3. Data extracted

Data extraction was conducted by two independent authors (K.P.O and E.R.O). Extracted data included study, demographic, clinical outcome, and complication data. Study information included first author, institution, year article was published, enrollment years, type of study, and level of evidence. Demographic data included cohort size, age, sex distribution, time until treatment, flexible fixation type, and follow-up duration. Clinical data included PROM scores such as the American Orthopedic Foot and Ankle Society (AOFAS), Visual Analog Scale (VAS), Maryland, Foot Function Index (FFI), 36-Item Short Form Survey (SF-36), Musculoskeletal Function Assessment (MFA), Short Musculoskeletal Function Assessment (SMFA), Veterans RAND 12-item (VR-12), and Foot and Ankle Ability Measure (FAAM) scores. [12] RTA and complications were recorded. Removal of hardware were excluded as complications as it was difficult to determine whether patients were symptomatic.

Table 1
MINORs criteria for quality assessment for flexible fixation studies.

Author	Clear Aim	Consecutive Patients	Prospective Data	Appropriate Endpoints	Unbiased assessment of study endpoint	Follow-up duration	Loss to follow-up less than 5 %	Total Score
Charlton	2	2	0	2	2	2	2	12
Cho	2	2	0	2	2	2	2	12
Cottom	2	2	0	2	2	2	2	12
Gee	2	2	0	2	2	2	2	12
Hoskins	2	2	0	2	2	2	2	12
Jain	2	2	0	2	2	2	2	12
Nery	2	2	0	2	2	2	2	12
Saxena	2	2	0	2	2	2	2	12
Sullivan	2	2	0	2	2	2	2	12
Yongfei	2	2	0	2	2	2	2	12

Table 2
MINORs criteria for quality assessment for ORIF/PA studies.

Author	Clear Aim	Consecutive Patients	Prospective Data	Appropriate Endpoints	Unbiased assessment of study endpoint	Follow-up duration	Loss to follow-up less than 5 %	Total Score
Abbasian	2	2	0	2	2	2	0	10
Cho	2	2	0	2	2	2	2	12
Gee	2	2	0	2	2	2	2	12
Ghate	2	2	0	2	2	2	2	12
Holyl	2	2	2	2	2	2	2	14
Huyke	2	2	0	2	2	2	0	10
Kandil	2	2	2	2	2	2	0	12
Korres	2	2	0	2	2	2	0	10
Kuo	2	2	0	2	2	2	2	12
Ly	2	2	2	2	2	2	2	14
Rajapakse	2	2	0	2	2	2	0	10
Reinhardt	2	2	2	2	2	2	2	14
Stodle	2	2	2	2	2	2	2	14
Urdahl	2	2	0	2	2	2	2	12
Vopat	2	2	0	2	2	2	2	12

2.4. Quality and bias assessment

Study qualities were analyzed using the Methodological Index for Non-Randomized Studies (MINORS).[13] The Grading of Recommendations, Assessment, Development and Evaluations (GRADE) criteria determine data certainty.

2.5. Statistical analysis

Statistical analysis was performed using SPSS Version 30 (IBM Corp., Armonk, NY). As there were only two comparative studies, meta-analyses were conducted using comparable FF and ORIF/PA groups from single-arm studies. The random effects model, as described by DerSimonian and Laird, was used to pool these studies. [14] When appropriate, ranges were converted to standard deviations. [15] If no range or standard deviation was present, the mean from the other studies was imputed. [16] Cohen's d reported the effect size of continuous data and odds ratios were used to report the effect size of binary data. Effect sizes were reported using Cohen's d (continuous) and odds ratio (OR, binary). Cohen's d values ≥ 0.2, ≥ 0.5, and ≥ 0.8 demonstrate a small, moderate, and large effects, respectively. [17] 95 % confidence intervals (CI) were reported. P-values ≤ 0.05 were statistically significant.

3. Results

3.1. Study background information and cohort characteristics

The database search generated 419 studies. After duplicate removal, 190 abstracts remained. After abstracts were screened, 47 full texts remained. After full texts were screened, 23 studies were included (two had flexible fixation and ORIF cohorts, eight studies reported FF results, and thirteen studies reported ORIF/PA results).[8,9,18–38] For studies that reported ORIF and/or PA, ten studies reported ORIF results, one study reported PA results, and four studies reported ORIF and PA results separately. (PRISMA Diagram, Fig. 1). All studies were sufficient quality according to MINORS criteria (Flexible Fixation Table 1, ORIF/PA Table 2).

Table 3
Study information for flexible fixation studies.

Author	Institution	Year Published	Years of Enrollment	Level of Evidence	Study Population
Charlton	Cedars Sinai, Los Angeles	2015	NA	IV	College soccer players and ballet dancers
Cho	Seoul	2021	2014 –2017	III	Sports injury
Cottom	Florida Orthopedic	2020	2008 –2016	III	All with injury
Gee	Tripler Army, Honolulu	2019	2009 –2012	IV	Armed Forces, Athletes
Hoskins	Penn State UC Davis	2024	2019 –2022	III	All with injury
Jain	Wigan, UK	2017	NA	IV	Athletes
Nery	Brazil	2019	NA	IV	All with injury
Saxena	Palo Alto California	2021	2008 –2016	IV	Athletes
Sullivan	Sydney, Australia	2022	2017 –2022	IV	Athletes
Yongfei	China	2021	2016 –2019	IV	All with injury

Table 4
Demographic information for flexible fixation studies.

Author	Patients, n	Age, years	Male, n	Time until treatment	Type of Flexible Fixation	Follow-up, y
Charlton	7	24.6 (18 –29)	1	14 (6 –20) months	TightRope, Arthrex	2.1 (1.3 –3.8)
Cho	31	40.9 (20 –69)	18	6.2 (2.16) DAYS	TightRope, Arthrex	1.3 (1 –2.2)
Cottom	84	37.7 (17 –72)	50	≤ 14 days	TightRope, Arthrex	3.4 (3 –5)
Gee	6	29.7	5	NA	TightRope, Arthrex	1.0 (0.46 –2.50)
Hoskins	15	35.2 (18 –66)	8	NA	TightRope, Arthrex	0.6 (0.12 –1.54)
Jain	5	22.1 (19 –27)	5	NA	TightRope, Arthrex	2 for all
Nery	7	36 (15 –55)	4	NA	TightRope, Arthrex	0.5 (0.25 –0.83)
Saxena	6	19.7 (14 –30)	2	NA	Endo-button	2.3 (0.83 –4)
Sullivan	12	21.1 (16 –34)	9	5.3 weeks (5.3)	TightRope, Arthrex	3.5
Yongfei	11	35.4 (16 –53)	8	4.5 (3 –8) days	TightRope, Arthrex	1.7 (1.4 –2)

Table 5
Patient-reported outcome measure scores for flexible fixation studies.

Author	PROM	PROM
Charlton	AOFAS	Pre: 65 (59 –72)
		Post 97 (90 –100)
	Return to activity	7/7 (at 6-months)
Cho	AOFAS	Pre: 45.1 (10.1)
		Post: 84.3 (7.8)
	VAS Pain	Pre: 7.9 (2.58)
		Post: 2.3 (1.61)
Cottom	AOFAS	Pre: 30.96 (25.75)
		Post: 90.36 (12.18)
	VAS Pain	Pre: 8.41 (1.73)
		Post: 1.3 (1.57)
Gee	Return to activity	6/6 (at 6-months)
	VAS pain	Post: 1.6
Hoskins	Return to activity	15/15 (14.1 weeks (11.6 –24))
Jain	Return to sport	5/5 (20.4 weeks (18 –24))
	AOFAS	Post: 94
	VAS Pain	Post: 0.6
Nery	Return to activity	7/7 (mean 6 mo (3 –10))
	AOFAS	Post: 92 (89 –98)
Saxena	Return to activity	6/6 (27 weeks (10 –48)
Sullivan	Return to sport	12/12 (12.7 weeks (1.7)
Youngfei	VAS Pain	Pre: 7.1 (1.0)
		Post: 1.5 (0.7)
	AOFAS	Post: 92.4 (4.3)
	Maryland Score	Post: 94.1 (3.5)

Table 6
Complication for flexible fixation studies.

Author	Complications	Details
Charlton	0	
Cho	2	2 Implant subsidence of medial cuneiform round button with bone erosion (65 yo and 67 yo)
Cottom	0	None
Gee	0	None
Hoskins	2	Medial bursitis near medial interference screw, hypersensitivity of foot > 6 weeks postoperatively.
Jain	1	Transient deep peroneal nerve sensory deficit
Nery	0	
Saxena	0	
Sullivan	2	Mild pain from medial button, transient paresthesia of dorsum of the foot
Yougfei	0	

3.2. Systematic review – flexible fixation

Ten studies that reported PROMs after FF. [8,9,18–25] Level of evidence was III for three studies and IV for seven studies. Patient enrollment was between 2008 and 2022 and publication was between 2015 and 2024. Six studies reported outcomes for athletes and four studies reported all injuries. Five (50 %) studies were located within the United States (Table 3). There were 184 patients across these studies. Mean age was 35.0 years (range, 19.7–40.9). Males comprised 59.8 % (n = 110/184) of the cohort. Five studies reported time from injury to treatment. Three studies had time to treatment of ≤ 14 days, one study had a mean time of 5.3 weeks, and one study had a mean time of 14 months. For fixation, TightRope (Arthrex, Inc., Naples, FL) was in 9/10 studies and EndoButton (Smith&Nephew, Inc., Memphis, TN) was used in the final study. Mean follow-up duration was 2.4 years (Table 4).

All studies reported PROM improvement (or high post-operative PROMs if pre-operative values could not be obtained) or return to activity after treatment. AOFAS scores were reported in six studies, VAS-pain scores were reported in five studies, the Maryland score was reported in one study, and return to activity rates were reported in seven studies. For AOFAS, 3/6 studies reported both pre- and post-operative scores and the remaining three studies only reported post-operative scores. In the three studies that reported both scores, there was a substantial improvement from 36.5 ± 20.8 to 89.2 ± 10.8 after treatment. Pooled post-operative AOFAS score was 89.7 ± 10.0. For VAS pain scores, three studies reported both pre- and post-operative scores which demonstrated substantial improvement after treatment from 8.2 ± 1.9 to 1.6 ± 1.5. Across the five studies reporting postoperative VAS-pain scores, mean score was 1.5 ± 1.5. RTA rate was 100 % (n = 58/58) (Table 5).

There were no reports of post-traumatic arthritis. Complication rate was 3.8 % (n = 7/184) and there were no reoperations pertaining to the suture button. Two complications were in seniors (65 and 67 years old) and were from subsidence of the medial cuneiform suture button. There was also medial bursitis near the cuneiform suture button, hypersensitivity of the foot > 6 weeks postoperatively, transient deep peroneal nerve sensory deficit, mild pain from the medial suture button, and transient paresthesia of the dorsum of the foot (Table 6). There were no cases of hardware removal, loss of reduction, or infection.

3.3. Systematic review – ORIF

Fourteen studies reported PROMs after ORIF. [8,9,26–34,36–38] Level of evidence was I for two studies, III for six studies, and IV for six studies. Patient enrollment was between 1997 and 2022 and publication was between 2000 and 2023. Two studies reported outcomes for athletes and twelve studies reported all injuries. Four (29 %) studies were located within the United States (Table 7). There were 236 patients across these studies. Mean age was 36.0 years (range, 24.6–56.8). Males comprised 67.7 % (n = 153/226) of the cohort. Time to treatment was < 6 h to 15 days. Mean follow-up duration was 3.7 years (Table 8).

Postoperatively, AOFAS scores, VAS pain scores, and RTA were the most reported PROMs. Ten studies reported AOFAS scores, four studies reported VAS pain scores, and five studies reported RTA. At the final follow-up evaluation, the AOFAS score was 78.7 ± 44.2, the VAS-pain score was 1.6 ± 3.8, and the return to activity rate was 63.3 % (n = 50/79). Other PROMs included FFI, SF-36, Maryland, SMFA, MFA, VR-12, and FAAM scores. Each of these scores improved after treatment (Table 9). Post-traumatic arthritis rate was 13.0 % (n = 28/216). Hardware removal rate was 86.0 % (n = 203/236). Complication rate was 17.7 % (n = 34/192). There were six cases of loss of reduction and nine cases of infection (Table 10).

Table 7
Study information for ORIF/PA studies.

Author	Institution	Year Published	Years of Enrollment	Level of Evidence	Study Population
Abbasian	Switzerland	2015	1998 –2012	III	All Injuries
Cho	Seoul	2021	2014 –2017	III	Athletes
Gee	Tripler Army, Honolulu	2019	2009 –2012	IV	Armed Forces, Athletes
Ghate	India	2012	2007 –2011	III	All Injuries
Holyl	Egypt	2021	2021	III	All Injuries
Huyke	Minneapolis	2023	2007 –2013	III	All Injuries
Kandil	Egypt	2022	2013 –2018	III	All Injuries
Korres	Greece	2003	1990 –2001	IV	All Injuries
Kuo	Harborview, Seattle	2000	1990 –1997	IV	All Injuries
Ly	Minneapolis	2006	1998 –2002	I	All Injuries
Rajapakse	New Zealand	2006	1997 –2001	IV	All Injuries
Reinhardt	New York	2012	1998 –2006	III	All Injuries
Stodle	Norway	2020	2011 –2015	I	All Injuries
Urdahl	Minneapolis	2023	2012 –2022	IV	All Injuries
Vopat	Netherlands	2019	2010 –2014	IV	Athletes

Table 8
Demographic information for ORIF/PA studies.

Author	Patients, n	Age, years	Male, n	Time until treatment	Rigid ORIF or PA	Follow-up, y
Abbasian	29	39.9 (17 –88)	19	NA	ORIF	8.3 (1 –15)
Cho	32	40.9 (20 –69)	21	5.8 (2.41)	ORIF	1.3 (1 –2.2)
Gee	6	25.7	5	NA	ORIF	1.0 (0.46 –2.50)
Ghate	6	43.3 (8.5)	4	< 10 days	ORIF	2.5 (0.3)
Holyl	18	56.77 (8.17)	4	NA	ORIF	1 min
Huyke	12	41.0 (13.5)	5	9.1 (5.3) days	ORIF	4.0 (1.7)
Kandil	16	44 ± 4.56	11	NA	PA	2.8 (2.2 –4.2)
	14	35 ± 3.45	8		ORIF	
Korres	6	24.7 (5.4)	6	< 6 h	ORIF	3.7 (1 –9)
Kuo	18	36.8 (16.5)	12	NA	ORIF	4.3 (1.1 –9.5)
Ly	21	32 (19 –42)	14	NA	PA	3.6 (2.1 –5)
	20	32.4 (19 –52)	13		ORIF	3.5 (2.1 –5)
Rajapakse	9	33.2 (16 –76)	6	NA	ORIF	3.6 (1 –5.8)
Reinhardt	12	51.5 (20 –71) *	4	NA	PA	3.5 (2 –8)
Stodle	24	30 (23 –40)	11	16.5 d	PA	2 –3 weeks
	24	34 (28 –40)	11	15 d	ORIF	6 –8 weeks
						12 weeks
						6 months
						1 year
						2 years
Urdahl	7	24.6 (14.3 –54.8)	4	NA	PA	3.7 (1 –7.7)
	32		19		ORIF	
Vopat	10	25.4 (15 –37)	NA	NA	ORIF	9.1 (1.2 –5)

3.4. Systematic review – primary arthrodesis

Five studies that reported PROMs after PA. [30,33,35–37] Level of evidence was I for two studies, III for two studies, and IV for one study. Patient enrollment was between 1998 and 2022 and publication was between 2006 and 2023. Three (60 %) studies were located within the United States (Table 7). There were 80 patients across these studies. Mean age was 36.1 years (range, 24.6–51.5). Males comprised 55.0 % (n = 44/80) of the cohort. Only one study reported time to treatment which was 16.5 days. Mean follow-up duration was 3.0 years (Table 8).

Postoperatively, AOFAS scores, VAS pain scores, and RTA were the most reported PROMs. Four studies reported AOFAS scores, two studies reported VAS pain scores, and two studies reported RTA. At the final follow-up evaluation, the AOFAS score was 87.4 ± 31.8, the VAS-pain score was 0.3 ± 2.6, and the RTA rate was 78.4 % (n = 29/37). Other PROMs included FFI, SF-36, SMFA, VR-12, and FAAM scores. All scores improved after treatment (Table 9). No patients developed post-traumatic arthritis. Complication rate was 23.5 % (n = 16/68). Hardware removal rate was 22.5 % (n = 18/80). There were no reports of loss of reduction and only one infection (Table 10).

3.5. Comparisons between flexible fixation / ORIF / PA

PROMs were reported for FF, ORIF, and PA. Postoperative AOFAS scores were 89.7 ± 10.0, 78.7 ± 44.2, and 87.4 ± 31.8, respectively. Postoperative VAS pain scores were 1.5 ± 1.5, 1.6 ± 3.8, and 0.3 ± 2.6, respectively. RTA rates were 100 %, 63.3 %, and 78.4 %, respectively. Prevalence of post-traumatic arthritis was 0 %, 13.0 %, and 0 %, respectively. Hardware removal rates were 0 %, 86.0 %, and 22.5 %, respectively. Complication rates were 3.8 %, 17.7 %, and 23.5 %, respectively (Table 11).

As FF is relatively newer, there are limited papers directly comparing FF to ORIF and PA. To preliminarily analyze the available data, pooled statistics were used to conduct a meta-analysis to compare FF, ORIF, and PA. Therefore, we conducted three meta-analyses—one compared FF to both ORIF and PA, one compared FF to ORIF, and one compared FF to PA.

The first meta-analysis compared FF to a combined ORIF/PA group. For AOFAS scores, FF demonstrated weak superiority over ORIF/PA with an effect size of 0.259 (CI:0.054–0.465, p = 0.013). For VAS-pain scores, neither treatment demonstrated superiority (p = 0.373). For RTA, patients were 58 times more likely to return after FF (OR:58.09, CI:3.49–967.36, p = 0.005). For post-traumatic arthritis, ORIF or PA had a 41 times increased chance for developing arthritis (OR:41.00,

Table 9
Patient-reported outcome measure scores for ORIF/PA studies.

Author	PROM	PROM
Abbasian	AOFAS	Post: 84 (27 –100)
	FFI Function	Post: 11.6 (0 –45)
	FFI Pain	Post: 9.9 (0 –36)
	SF36 Physical	Post: 56.2 (32 –100)
	SF36 Mental	Post: 57 (23 –100)
	VAS Pain	1.6 (0 –5)
	Return to activity	22/29
Cho	AOFAS	Pre: 44.5 (6.92) Post: 86.2 (2.25)
	VAS Pain	Pre: 8.3 (2.62) Post: 2.1 (1.25)
Gee	Return to full duty	3/6 (358 days)
	VAS Pain	2.1
Ghate	AOFAS	Post: 73.5 (4.7)
	Maryland	Post: 72.2 (5.1)
Holyl	AOFAS	Post: 46.55 (4.74) 3 Month: 73.55 (6.55) 6 Month: 84.77 (3.87) 12 Month: 92.61 (3.95)
Huyke	SMFA Daily Activity	Post: 7.5 (9.2)
	SMFA Emotional	Post: 17.9 (12.3)
	SMFA Arm/ Hand	Post: 0.26 (0.9)
	SMFA Mobility	Post: 13.7 (12.9)
	SMFA Dysfunction Index	Post: 9.6 (7.9)
	SMFA Bothersome Index	Post: 13.7 (9.8)
Kandil	AOFAS	88.9 (60 –100) 61.7 (30 –90)
	FFI	22.7 ± 3.6 34.5 ± 4.5
	Return to previous level of activity, time to return	14 (89); TTR: 19 ± 3.5 11 (66); TTR: 17 ± 2.6
Korres	AOFAS	64.16 (3.43)
Kuo	AOFAS	77 (40 –100)
	MFA	19 (0 –55)
Ly	AOFAS	PA: 86.9 (63 –100) ORIF: 57.1 (16 –100)
	Return to previous level of activity, time to return	PA: 15 (71) ORIF: 6 (30)
Rajapakse	AOFAS	80.9 (38 –100)
Reinhardt	AOFAS	83.3 (49 –100)
	SF–36 Physical	48.1 (9.5)
	SF–36 Mental	54.9 (8.0)
	% of pre-injury level	86.9 %
	VAS Pain	1.5 (0 –4)
Stodle	AOFAS	PA: 89 ± 9 ORIF: 85 ± 15
	VAS Pain	PA: 0 (0 –2) ORIF: 0.9 (0 –3)
	SF36 PF	PA: 95 (89 –100) ORIF: 95 (89 –100)
	SF36 Role Physical	PA: 100 (44 –100) ORIF: 100 (88 –100)
Urdahl (PA and ORIF)	VR–12 Mental	Pre: 53.8 (37.5 –66) Post: 60.7 (48.1 –66.8
	VR–12 Physical	Pre: 34.7 (24.3 –44.1) Post: 51.8 (31.7 –57.6)
	FAAM Total	Pre: 30.5 (4.3 –55.2) Post: 91.1 (56.0 –100)
	FAAM ADL	Pre: 34.9 (6 –61.9) Post: 92.9 (66.7 –100)
	FAAM Sports	Pre: 20 (0 –39.3) Post: 86.5 (28.1 –100)
	VAS	Pre: 51.7 (4.1 –87.4) Post: 11.1 (0 –60.5)
Urdahl (ORIF only)	VR–12 Mental	Pre: 53.2 (37.5 –66) Post: 61 (48.1 –66.8)
	VR–12 Physical	Pre: 34.1 (24.3 –44.1) Post: 53.9 (45.8 –57.5)
	FAAM Total	Pre: 28 (4.3 –55.2) Post: 95.6 (81.9 –100)
	FAAM ADL	Pre: 32.6 (6 –61.9) Post: 96.7 (83.3 –100)
	FAAM Sports	Pre: 17.2 (0 –39.3) Post: 92.8 (78.1 –100)
	VAS	Pre: 55.2 (17.5 –87.4) Post: 11.3 (0 –35)
Vopat	Return to training	10/10 (27.1 weeks (16 –56))
	Return to competition	8/10 (29.4 weeks (22 –52)
	% of preinjury level	87 % (70 –100)

CI:2.49–675.86, p = 0.009). For hardware removal, ORIF or PA had a very large increased chance of undergoing future hardware removal (OR:855.85, CI:52.78–13877.24, p < 0.001). For complication rate, ORIF or PA had a 6 times increased chance of having a complication (OR:6.02, CI:2.66–13.61, p < 0.001) (Table 12). The number needed to treat (NNT) to prevent a complication when comparing FF to ORIF/PA was 7.

The second meta-analysis compared FF to ORIF. For AOFAS scores, FF demonstrated weak superiority over ORIF with an effect size of 0.331 (CI:0.110–0.552, p = 0.003). For VAS-pain scores, neither treatment demonstrated superiority (p = 0.804). For RTA, patients were 68 times more likely to return after FF (OR:68.35, CI:4.07–1147.14, p = 0.003). For post-traumatic arthritis, ORIF had a 56 times increased chance for developing arthritis (OR:55.79, CI:3.38–920.56, p = 0.005). For hardware removal, ORIF had a very large increased chance of undergoing future hardware removal (OR:2241.54, CI:136.37–36843.43, p < 0.001). For complication rate, ORIF had a 5 times increased chance of having a complication (OR:5.44, CI:2.35–12.62, p < 0.001) (Table 13).

The final meta-analysis compared FF to PA. For AOFAS scores, neither treatment demonstrated superiority (p = 0.425). For VAS-pain scores, PA demonstrated moderate superiority over FF with an effect size of 0.697 (CI:0.322–1.071, p < 0.001). For RTA, patients were 34 times more likely to return after FF (OR:33.71, CI:1.88–604.35, p = 0.017). For post-traumatic arthritis, PA had no effect on development of arthritis compared to FF (p = 1.00). For hardware removal, PA had a very large increased chance of undergoing future hardware removal (OR: 109.22, 95 % CI: 6.49–1839.09, p = 0.001). For complication rate, PA had an 8 times increased chance of having a complication compared to FF (OR: 7.78, 95 % CI: 30.4–19.93, p < 0.001) (Table 14).

There were two studies that directly compared FF to ORIF and there was no clear superior treatment between these treatments.[8,9] Cho et al. reported 31 FF patients and 32 ORIF patients.[8] Postoperative AOFAS scores were 84.3 ± 7.8 and 86.2 ± 2.3, respectively. Postoperative VAS-pain scores were 2.3 ± 1.6 and 2.1 ± 1.3, respectively. These differences are likely not clinically significant. Gee et al. reported on 6 FF patients and 6 ORIF patients.[9] Postoperative VAS-pain scores were 1.6 and 2.1, respectively. Return to activity rates were 100 % (n = 6/6) and 50 % (n = 3/6), respectively.

3.6. Level of certainty of estimates (GRADE)

The GRADE guidelines determined estimate certainties. All outcomes (AOFAS, VAS-pain, RTA, post-traumatic arthritis, hardware removal, complications) were low-quality based on the literature available during this review (Table 15).

4. Discussion

Injury of the Lisfranc ligament causes diastasis between the first and second metatarsals which leads to foot pain and difficulty with weight bearing if left untreated. [2] Historically, treatment was limited to open reduction internal fixation (ORIF) or primary arthrodesis (PA); however, these treatments do not maintain the native biomechanics due to rigid fixation across this joint. [3] Therefore, flexible fixation was introduced to allow for healing while allowing movement across this joint. [6]

Table 10
Post-traumatic arthritis and complications.

Author	Post-Traumatic Arthritis	Complication	Details	Reoperation
Abbasian	5 mild 3 moderate	3	3 loss of reduction	1 secondary arthrodesis
Cho	2	4	4 screw breakages (1 diastasis)	
Gee	1	1	1 loss of reduction	1 secondary arthrodesis
Ghate	2	NA	NA	0
Holyl	0	7	4 Infection, 1 Transient Numbness, 1 Delayed Healing, 1 Loss of Reduction	0
Huyke	0	0	None	0
Kandil	0 5	PA: 10 ORIF: 9	5 sympathetic dystrophy, 5 wound complications 4 sympathetic dystrophy, 5 wound complications	4 secondary arthrodesis
Korres	4	0	None	0
Kuo	6	NA	NA	0
Ly	PA: 0 NA	3 NA	1 delayed union, 1 non-union, 1 compartment syndrome resulting in claw toes NA	1 percutaneous flexor tendon release for claw toes, 1 redo arthrodesis 5 arthrodesis
Rajapakse	0	1	1 screw breakage	0
Reinhardt	NA	NA	Purely ligamentous complications could not be separated from osseous	
Stodle	PA: 0 ORIF: 0	2 6	1 superficial peroneal nerve injury, 1 nonunion ORIF: 1 superficial wound infection, 2 sympathetic dystrophy, 1 screw loosening leading to rupture of extensor hallucis longus tendon (required 2nd toe extensor transfer), 2 nonunion	2nd toe extensor transfer to great toe
Urdahl	0	2	1 persistent instability and 1 nonunion	2 revisions
Vopat	0	2	1 Infection, 1 DVT, 1 broken screw	0

While there are only two studies directly comparing FF and ORIF/PA, there are several more studies that evaluate one or the other cohorts. This systematic review and meta-analysis (SRMA) has attempted to synthesize that data and has demonstrated that all three treatment types have satisfactory postoperative PROMs, high return to activity, and low complication rates.

Postoperatively, PROM scores were satisfactory for FF, ORIF, and PA.

Table 11
Pooled data from FF, ORIF, and PA studies.

Parameter	Flexible Fixation	ORIF	Primary Arthrodesis
AOFAS, mean (SD)	89.7 (10.0)	78.7 (44.2)	87.4 (31.8)
VAS pain, mean (SD)	1.5 (1.5)	1.6 (3.8)	0.3 (2.6)
Return to Activity, % (n)	100 % (58/58)	63.3 % (50/79)	78.4 % (29/37)
Post-Traumatic Arthritis, % (n)	0 % (0/184)	13.0 % (28/216)	0 % (0/68)
Hardware Removal, % (n)	0 % (0/184)	86.0 % (203/236)	22.5 % (18/80)
Complications, % (n)	3.8 % (7/184)	17.7 % (34/192)	23.5 % (16/68)

Table 12
Results of meta-analysis for flexible fixation compared to combined ORIF / PA.

Parameter	Directionality of Effect	Effect Size	95 % CI	p-value
AOFAS	FF Superior	0.259	0.054 −0.465	0.013
RTA	FF Superior	58.09	3.49 −967.36	0.005
Post-traumatic Arthritis	ORIF/PA with Higher Prevalence	41.00	2.49 −675.86	0.009
Hardware Removal	ORIF/PA with Higher Prevalence	855.85	52.78 −13877.24	< 0.001
Complications	ORIF/PA with Higher Prevalence	6.02	2.66 −13.61	< 0.001

Effect size measured as Cohen's d for continuous measurements and as odds ratios (OR) for binary measurements.
Only significant values are reported.

Table 13
Results of meta-analysis for flexible fixation compared to combined ORIF.

Parameter	FF or ORIF Superior?	Effect Size	95 % CI	p-value
AOFAS	FF Superior	0.331	0.110 −0.552	0.003
RTA	FF Superior	68.35	4.07 −1147.14	0.003
Post-traumatic Arthritis	ORIF with Higher Prevalence	55.79	3.38 −920.56	0.005
Hardware Removal	ORIF with Higher Prevalence	2241.54	136.37 −36843.43	< 0.001
Complications	ORIF with Higher Prevalence	5.44	2.35 −12.62	< 0.001

Effect size measured as Cohen's d for continuous measurements and as odds ratios (OR) for binary measurements.
Only significant values are reported.

Table 14
Results of meta-analysis for flexible fixation compared to combined PA.

Parameter	FF or PA Superior?	Effect Size	95 % CI	p-value
VAS Pain	PA Superior	0.697	0.322 −1.071	< 0.001
RTA	FF Superior	33.71	1.88 −604.35	0.017
Hardware Removal	ORIF/PA with Higher Prevalence	109.22	6.49 −1839.09	0.001
Complications	ORIF/PA with Higher Prevalence	7.78	30.4 −19.93	< 0.001

Effect size measured as Cohen's d for continuous measurements and as odds ratios (OR) for binary measurements.
Only significant values are reported.

Table 15
GRADE assessment.

MA Outcome	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Effect Size	Final Assessment
AOFAS	-	-	-	+	+	-	Low
VAS Pain	-	-	-	+	+	-	Low
RTA	-	-	-	+	+	-	Low
Post-traumatic Arthritis	-	-	-	+	+	-	Low
Hardware Removal	-	-	-	+	+	-	Low
Complications	-	-	-	+	+	-	Low

AOFAS scores were 89.7 ± 10.0, 78.7 ± 44.2, and 87.4 ± 31.8, respectively. VAS pain scores were 1.5 ± 1.5, 1.6 ± 3.8, and 0.3 ± 2.6, respectively. For the other, non-overlapping PROMs, such as the Maryland score and short form scores, there were satisfactory postoperative scores as well. Return to function rates were 100 %, 63.3 %, and 78.4 %, respectively. To determine significance, we conducted meta-analyses by pooling data from the studies to determine whether AOFAS scores, VAS pain scores, or RTA were statistically different between groups. First, we compared FF to a combined ORIF/PA group. FF was superior to conventional treatment (ORIF and PA) in terms of AOFAS scores ($p = 0.013$) and RTA ($p = 0.005$). VAS pain scores did not demonstrate a statistical difference ($p = 0.373$). Next, we compared FF to ORIF. FF was superior to ORIF in terms of AOFAS scores ($p = 0.003$) and RTA ($p = 0.003$). VAS pain scores did not demonstrate a statistical difference ($p = 0.804$). Lastly, we compared FF to PA. FF was superior to PA in terms of RTA ($p = 0.017$) and PA was superior to FF in terms of VAS pain scores ($p < 0.001$). AOFAS scores did not demonstrate a statistical difference ($p = 0.425$).

This SRMA also recorded rates of development of post-traumatic arthritis, hardware removal, and complications. We compared FF, ORIF, and PA. Prevalence of post-traumatic arthritis at final follow-up was 0 %, 13.0 %, and 0 %, respectively. Hardware removal rates were 0 %, 86.0 %, and 22.5 %, respectively. Complication rates were 3.8 %, 17.7 %, and 23.5 %, respectively. Once again, we compared FF to conventional treatment (ORIF and PA), ORIF alone, and PA alone. First, we compared FF to conventional treatment. Undergoing ORIF or PA had a higher likelihood of having post-traumatic arthritis ($p = 0.009$), undergoing a hardware removal ($p < 0.001$), or having complications ($p = 0.004$). Next, we compared FF to ORIF. Undergoing ORIF had a higher likelihood of having post-traumatic arthritis ($p = 0.005$), undergoing a hardware removal ($p < 0.001$), or having complications ($p < 0.001$). Lastly, we compared FF to PA. Undergoing PA had no effect on development of post-traumatic arthritis ($p = 1.00$); however, it did have a higher likelihood of undergoing a hardware removal ($p = 0.001$) or having complications ($p < 0.001$).

The two studies that compared FF to ORIF (Cho et al. and Gee et al.) did not demonstrate a preference for treatment type. [8,9] A pooled meta-analysis was performed which demonstrated a weak preference for FF using AOFAS scores, no preference using VAS pain scores, a strong preference for FF for return to activity and decreased complication rates. Utilizing all available data for AOFAS scores, FF had outcomes 8.5 points higher than the conventional treatment (rigid fixation). Most of this came from the superiority of FF over ORIF. In a recent study on hallux valgus surgery, Chan et al. demonstrated that the minimal clinically important difference (MCID) in AOFAS scores was 7.9. Although the MCID for AOFAS scores in hallux valgus surgery may not directly correlate to Lisfranc injury, this data is the closest estimate we have for an MCID because there is no currently published MCID for AOFAS scores in Lisfranc injury. By the best estimate, therefore, our pooled data demonstrated improved clinical outcomes for FF over the conventional treatment for Lisfranc injury [39].

Financial considerations may play a factor in treatment/implant choice for this injury. The additional cost associated with hardware removal may be relevant in the case of rigid fixation when compared to flexible fixation. There is no published literature on cost of treatment for Lisfranc hardware removal in comparison to costs of flexible fixation. Lalli et al. found that the cost of syndesmotic screw hardware removal in the ankle was on average of $3579 in 2015. [40] Given the rate of hardware removal with rigid fixation of Lisfranc injuries (86 % in ORIF), hardware removal causes a significant economic impact that likely exceeds the additional cost of flexible implants, similar to the economic analysis seen in treatment of syndesmotic injuries. [41] Furthermore, undergoing an extra surgical procedure may cause additional patient discomfort, time off work, and inconvenience.

The current data would suggest that FF is likely equally effective and safe in treating ligamentous Lisfranc injuries compared to ORIF and PA due to the high PROMs and return to activity rates. However, these studies are limited in their sample size, are retrospective, and mostly single arm. The literature would benefit from further comparative studies and randomized controlled trials. Also, there are reporting biases when discussing new treatments strategies that must be understood. For example, while the reported return to activity is 100 % for FF, there are likely some patients that will be unable to return to activity with FF. This SRMA is limited to low-quality evidence; however, it does summarize and attempt to analyze the most up-to-date evidence regarding FF.

While the current paper summarizes all available literature to compare FF to ORIF/PA, there are limitations. As these studies are single arm, there is no objective way to generate funnel plots or investigate for publication bias without utilizing assumptions. Furthermore, the heterogeneity of reporting PROMs makes analysis difficult across studies that do not utilize similar PROMs. For FF, there is a paucity of clinical outcomes research and development of prospective, randomized trials would improve the understanding of its use in Lisfranc injuries. We recommend using AOFAS scores as the primary endpoint when designing such trials in order to allow better comparisons between treatments and across studies in the future. Utilizing our mean AOFAS scores for FF (89.7) and ORIF/PA (81.2), we calculated that only 22 patients in each group (44 total) would be needed for 80 % power.

5. Conclusion

This review demonstrated that FF for Lisfranc injuries have high post-operative AOFAS and VAS pain scores, high return to activity rates, and low rates of complications which were comparable to ORIF and PA. A pooled meta-analysis demonstrated weak preference for FF; however, further comparative studies are needed to investigate how FF compares to ORIF and PA.

CRediT authorship contribution statement

Kyle P O'Connor: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Erica R Olfson:** Data curation. **John T Riehl:** Writing – review & editing, Visualization, Validation, Supervision, Conceptualization.

Declaration of Competing Interest

JTR is currently a consultant and receiving royalties for Arthrex, Inc., Naples, Florida. For the remaining authors, none are declared.

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References

[1] de Palma L, Santucci A, Sabetta SP, Rapali S. Anatomy of the Lisfranc joint complex. Foot Ankle Int 1997;18:356–64.
[2] Welck MJ, Zinchenko R, Rudge B. Lisfranc injuries. Injury 2015;46:536–41.
[3] Watson TS, Shurnas PS, Denker J. Treatment of Lisfranc joint injury: current concepts. J Am Acad Orthop Surg 2010;18:718–28.
[4] Grewal US, Onubogu K, Southgate C, Dhinsa BS. Lisfranc injury: a review and simplified treatment algorithm. Foot (Edinb) 2020;45:101719.
[5] Mascio A, Greco T, Maccauro G, Perisano C. Lisfranc complex injuries management and treatment: current knowledge. Int J Physiol Pathophysiol Pharm 2022;14:161–70.
[6] Briceno J, Stupay KL, Moura B, Velasco B, Kwon JY. Flexible fixation for ligamentous lisfranc injuries. Injury 2019;50:2123–7.
[7] Chona DV, Park CN, Kim BI, Lau BC. Clinical and biomechanical outcomes of suture button fixation for ligamentous lisfranc injury: a systematic review and meta-analysis. Orthop J Sports Med 2023;11. 23259671231186387.
[8] Cho J, Kim J, Min TH, Chun DI, Won SH, Park S, et al. Suture button vs conventional screw fixation for isolated lisfranc ligament injuries. Foot Ankle Int 2021;42:598–608.
[9] Gee S, Harris MC, Anderson C, Groth A, Ryan P. Lisfranc open reduction and internal fixation in an athletic population: screw versus suture button fixation. Curr Orthop Pract 2019;30:323–6.
[10] Eriksen MB, Frandsen TF. The impact of patient, intervention, comparison, outcome (PICO) as a search strategy tool on literature search quality: a systematic review. J Med Libr Assoc 2018;106:420–31.
[11] Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. Bmj 2021;372:n71.
[12] Lakey E, Hunt KJ. Patient-reported outcomes in foot and ankle orthopedics. Foot Ankle Orthop 2019;4. 2473011419852930.
[13] Slim K, Nini E, Forestier D, Kwiatkowski F, Panis Y, Chipponi J. Methodological index for non-randomized studies (minors): development and validation of a new instrument. ANZ J Surg 2003;73:712–6.
[14] DerSimonian R, Laird N. Meta-analysis in clinical trials. Control Clin Trials 1986;7:177–88.
[15] Walter SD, Yao X. Effect sizes can be calculated for studies reporting ranges for outcome variables in systematic reviews. J Clin Epidemiol 2007;60:849–52.
[16] Furukawa TA, Barbui C, Cipriani A, Brambilla P, Watanabe N. Imputing missing standard deviations in meta-analyses can provide accurate results. J Clin Epidemiol 2006;59:7–10.
[17] Sullivan GM, Feinn R. Using effect size-or why the P value is not enough. J Grad Med Educ 2012;4:279–82.
[18] Charlton T, Boe C, Thordarson DB. Suture button fixation treatment of chronic lisfranc injury in professional dancers and high-level athletes. J Dance Med Sci 2015;19:135–9.
[19] Cottom JM, Graney CT, Sisovsky C. Treatment of lisfranc injuries using interosseous suture button: a retrospective review of 84 cases with a minimum 3-year follow-up. J Foot Ankle Surg 2020;59:1139–43.
[20] Hoskins M, Wise P, Unangst A, Shaheen P, Kreulen C, Aynardi M, et al. Early outcomes of lisfranc injuries treated with arthrex internalBrace: a case series. Indian J Orthop 2024;58:257–62.
[21] Jain K, Drampalos E, Clough TM. Results of suture button fixation with targeting device aid for displaced ligamentous Lisfranc injuries in the elite athlete. Foot (Edinb) 2017;30:43–6.
[22] Nery C, Giza E, Wagner E, Percope de Andrade MA, Baumfeld TS, Prado MP, et al. Dynamic lisfranc joint repair concept: Surgical technique for a synthetic neoligamentplasty. Muscles Liga Tendons J 2019;9:562–70.
[23] Saxena A, Shou L, Fournier M. Results of lisfranc's surgery in athletic patients. Muscles Liga Tendons J 2021;11:736–41.
[24] Sullivan M, Peckston D, Alpuerto B. 2nd. Shortening the return-to-play times in elite athletes with unstable isolated lisfranc ligament injuries using a knotless interosseous suture button: case series and literature review. Orthop J Sports Med 2022;10:23259671221102969.
[25] Yongfei F, Chaoyu L, Wenqiang X, Xiulin M, Jian X, Wei W. Clinical outcomes of Tightrope system in the treatment of purely ligamentous Lisfranc injuries. BMC Surg 2021;21:395.
[26] Abbasian MR, Paradies F, Weber M, Krause F. Temporary internal fixation for ligamentous and osseous lisfranc injuries. Foot Ankle Int 2015;36:976–83.
[27] Ghate SD, Sistla VM, Nemade V, Vibhute D, Shahane SM, Samant AD. Screw and wire fixation for Lisfranc fracture dislocations. J Orthop Surg (Hong Kong) 2012;20:170–5.
[28] Holyl SM, Mohammed EA, Hadida AM, Imam AHF. Short term assessment of surgical treatment of Lisfranc fracture dislocation. Egypt J Hosp Med 2021;85:2945–50.
[29] Huyke-Hernández FA, Lesmeister N, Yonke B, Mazzuca J. CT and functional outcomes of primarily ligamentous and combined ligamentous-bony lisfranc injuries: a retrospective review. J Foot Ankle Surg 2023;62:317–22.
[30] Kandil MI, Abouzeid M, Eltaher SM, Eltregy S. Primary fusion versus open reduction internal fixation for purely ligamentous lisfranc injuries: A Prospective comparative study and analysis of factors affecting the outcomes. Foot Ankle Surg 2022;28:898–905.
[31] Korres DS, Psicharis IP, Gandaifis N, Papadopoulos EC, Zoubos AB, Nikolopoulos K. Outcome after anatomic reduction and transfixation with Kirschner wires of Lisfranc joint injuries. Eur J Orthop Surg Traumatol 2003;13:85–90.
[32] Kuo R, Tejwani N, DiGiovanni C, Holt S, Benirschke S, Hansen S, et al. Outcome after open reduction and internal fixation of lisfranc joint injuries*. J Bone Jt Surg Am 2000;82:1609.
[33] Ly TV, Coetzee JC. Treatment of primarily ligamentous Lisfranc joint injuries: primary arthrodesis compared with open reduction and internal fixation. A prospective, randomized study. J Bone Jt Surg Am 2006;88:514–20.
[34] Rajapakse B, Edwards A, Hong T. A single surgeon's experience of treatment of Lisfranc joint injuries. Injury 2006;37:914–21.
[35] Reinhardt KR, Oh LS, Schottel P, Roberts MM, Levine D. Treatment of Lisfranc fracture-dislocations with primary partial arthrodesis. Foot Ankle Int 2012;33:50–6.
[36] Stødle AH, Hvaal KH, Brøgger HM, Madsen JE, Husebye EE. Temporary bridge plating vs primary arthrodesis of the first tarsometatarsal joint in lisfranc injuries: randomized controlled trial. Foot Ankle Int 2020;41:901–10.
[37] Urdahl TH, Dock CC, Stone McGaver R, Seiffert KJ, Coetzee JC. Outcomes of surgically treated purely ligamentous stage II Lisfranc Injuries. Foot Ankle Int 2023;44:1120–7.
[38] Vopat BG, Vopat ML, van Dijk PAD, Hazzard S, McKinnon K, Asnis PD, et al. Return to sport after surgical treatment of lisfranc injuries in athletes: a retrospective case series. Kans J Med 2019;12:141–5.
[39] Chan HY, Chen JY, Zainul-Abidin S, Ying H, Koo K, Rikhraj IS. Minimal clinically important differences for american orthopaedic foot & ankle society score in hallux valgus surgery. Foot Ankle Int 2017;38:551–7.
[40] Lalli TA, Matthews LJ, Hanselman AE, Hubbard DF, Bramer MA, Santrock RD. Economic impact of syndesmosis hardware removal. Foot (Edinb) 2015;25(3):131.
[41] Weber AC, Hull MG, Johnson AJ, Henn RF. 3rd. Cost analysis of ankle syndesmosis internal fixation. J Clin Orthop Trauma 2019;10:173–7.